

On the power-series construction of Schrödinger bound states.

II. The effective Hill determinants

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For a class of potentials, the radial differential Schrödinger equation for wave functions $\psi(r)$ of bound states may often be replaced by an equivalent matrix equation for coefficients p_n in an associated power-series ansatz. In Paper I [J. Math. Phys. **29**, 1433 (1988)], a rigorous foundation of such a transition (sometimes called the Hill-determinant method) was presented. Now, it will be shown that and how the Feshbach–Löwdin projection-operator idea may help one to reduce the underlying infinite-dimensional diagonalization to a mere finite-dimensional eigenvalue problem.

I. INTRODUCTION

In the preceding paper (I),¹ we have considered the Schrödinger bound-state problem

$$\left[-\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} + V(r) \right] \psi(r) = E\psi(r),$$

$$\psi \in L_2(0, \infty), \quad l = 0, 1, \dots, \quad (1.1)$$

with a class of potentials equivalent, up to a change of variables, to the polynomials

$$V(r) = \sum_{n=1}^{2q+1} a_n r^{2n}, \quad a_{2q+1} = a^2 > 0. \quad (1.2)$$

$$Q = Q^{(N)} = \begin{pmatrix} A_0 & -B_0 & 0 & \cdots & \cdots & 0 \\ C_1^{(1)} & A_1 & -B_1 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & \cdots & 0 & D^{(q)} & D^{(q-1)} & \cdots D^{(1)} & C_N^{(q)} & \cdots & A_N \end{pmatrix} \quad (1.4)$$

[cf. Eqs. (5.3)–(5.5) in I].

In the literature,^{3–16} the whole approach is usually called the method of Hill determinants. It is based on the ansatz

$$\psi(r) = \exp(-g(r)) \sum_{n=0}^{\infty} p_n r^{2n+l+1}, \quad (1.5)$$

with some (polynomial) $g(r)$. An insertion of (1.5) in (1.1) leads immediately to an explicit form of matrix elements in Eq. (1.3). Vice versa, each numerical solution of the truncated Eq. (1.3) has been proposed^{3–6} to convert Eq. (1.5) into an approximate wave function, which becomes exact in the $N \rightarrow \infty$ limit.

Recently, the above Hill-determinant (HD) method has been criticized⁷ and shown to have a restricted validity,^{8–13} which, essentially, depends on our choice of $g(r)$. In I, the latter point has been clarified for all the potentials (1.2). We proposed a use of polynomials

$$g(r) = \frac{a}{2q+2} r^{2q+2} + \sum_{j=1}^q \frac{\beta_j}{2j+2} r^{2j},$$

$$a > 0, \quad a^2 = a_{2q+1}, \quad (1.6)$$

In essence, we have proved an equivalence of the differential equation (1.1) (with the standard boundary conditions²) to an algorithm based on an infinite-dimensional limiting transition in the $(N+1)$ -dimensional matrix equation

$$Qz = Ez, \quad z = \begin{pmatrix} p_0 \\ p_1 \\ \vdots \\ p_N \end{pmatrix}, \quad (1.3)$$

with a “quasi-Hamiltonian” matrix Q possessing $2q+2$ diagonals and Hessenberg structure,

which reflect the correct WKB $r \rightarrow \infty$ asymptotics of $\psi(r)$. All the possible failures (cf., e.g., their $q=1$ examples and analysis in Refs. 8–13) were eliminated by means of the simple restriction

$$a_{2q}/2a = \gamma_q > -\beta_q, \quad (1.7)$$

which has to be added to (1.6).

In the resulting well-founded form, the HD technique is of interest as a possible new type of resummation of the divergent perturbation series.⁴ The recent increase of interest in it also reflects its simplicity. Indeed, mere $2q+2$ diagonals in Q replace the $4q+3$ nonzero diagonals of the Hamiltonian when represented as a matrix in an ordinary harmonic oscillator basis.¹¹

Numerically, the latter variational diagonalization converges extremely quickly. Thus some authors also emphasize an improvement of efficiency of the purely numerical HD algorithms.^{10,12,14} The latter effort proved successful: The extreme simplicity of the matrix elements (1.4) enables one to reduce the infinite-dimensional (exact) HD secular equation

$$\det(Q - E) = 0 \quad (1.8)$$

to a mere transcendental equation

$$1/F_0 = 0, \quad (1.9)$$

where F_n denotes a certain “extended” analytic continued fraction⁶ (and coincides with the ordinary continued fraction in the simple example of Ref. 4). This arrangement makes full use of the specific Hessenberg structure of Q and simplifies significantly the numerical infinite-dimensional limiting transition $N \rightarrow \infty$.

Geometrically, the asymptotically dominant components of F_n , $n \gg 1$, may be identified with the so-called fixed points (FP’s) of the underlying continued-fractional mappings.⁶ Thus our algebra proceeds one step further and we can replace the recurrently defined quantities F_n , $n \gg 1$, by certain FP expansions in roots of algebraic equations of degree $q + 1$ (see Ref. 6).

The first numerical test of the above FP asymptotic expansion idea has already been performed in Ref. 14, for the potential $V(r) = r^2 + gr^4$ and with the first- and second-order FP corrections taken into account. The convergence still remains comparatively slow. Fortunately, an incorporation of the higher-order corrections makes the combined FP–HD algorithm fully competitive with the other methods. On a few examples, this was demonstrated numerically by Tater and co-workers.^{10,15}

An unpleasant shortcoming of the rather universal FP technique lies in a tedious algebra pertaining to the derivation of the explicit FP corrections. In a way, these complications stem from the “too geometric” character of the original

algorithms. In the present paper based on a close analogy between the Hermitian and non-Hermitian cases, we recall an experience gained in the former case¹⁷ and describe a simplified construction of the FP expansions.

The core of our contribution is formulated in Sec. II. In place of the $N = \infty$ eigenvalue condition (1.8), we propose use of the finite-dimensional rigorous formula

$$\det(Q^{\text{eff}} - E) = 0. \quad (1.10)$$

This will complement and complete the results of Refs. 1 and 6. In Sec. III, we change our point of view, reinterpret Eq. (1.3) as a difference equation, and derive a new formula for the unknown matrix elements of Q^{eff} . Finally, an illustration of technicalities is provided by Sec. IV where both the algebraic and numerical aspects of our eigenvalue condition (1.10) are discussed for the $q = 1$ example. Section V is a summary.

II. THE FINITE-DIMENSIONAL “EFFECTIVE” HILL DETERMINANTS

A. The extended continued fractions (ECF)

In the spirit of Ref. 6, we may postulate that

$$Q - E = U \cdot L, \quad (2.1)$$

where the matrix U has only two nonzero diagonals,

$$U_{nn} = 1, \quad U_{nn+1} = -B_n F_{n+1}, \quad n \geq 0, \quad (2.2)$$

while L is a lower triangular matrix with $2q + 1$ diagonals, in general:

$$L = \begin{pmatrix} Z_0 & 0 & 0 & \cdots \\ Y_1^{(1)} & Z_1 & 0 & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ 0 & \cdots & 0 & X_n^{(q)} & \cdots & X_n^{(1)} & Y_n^{(q)} & \cdots & Y_n^{(1)} & Z_n & 0 & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{pmatrix}. \quad (2.3)$$

The new matrix elements can be computed, from definition (2.1), by recurrences. Their general form may be found elsewhere¹⁴. For the present purposes, we restrict our attention to the first nontrivial example with $q = 1$ and, abbreviating

$$G_n = C_n^{(1)} + B_n F_{n+1} D^{(1)}, \quad n \geq 0, \quad (2.4)$$

write the definitions

$$Y_n^{(1)} = G_n, \quad X_n^{(1)} = D^{(1)}, \quad Z_n = 1/F_n,$$

and recurrences

$$F_n = 1/(A_n - E + B_n F_{n+1} G_{n+1}), \quad n = 0, 1, \dots \quad (2.5)$$

For $D^{(1)} = 0$, the latter ECF rule defines simply the ordinary continued fractions.⁴

B. The Löwdin–Feshbach projection operators in the HD non-Hermitian case

Let us assume that the $N \rightarrow \infty$ problem of convergence has been settled⁶ and fix some large dimension $N < \infty$. Next,

for reasons that will become obvious immediately, let us also introduce a pair of projectors $\hat{p}$ and $\hat{q} = 1 - \hat{p}$ such that

$$\hat{p}z = \hat{p} \begin{pmatrix} p_0 \\ p_1 \\ \vdots \\ p_N \end{pmatrix} = \begin{pmatrix} p_0 \\ p_1 \\ \vdots \\ p_M \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad (2.6)$$

$$\hat{q}z = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ p_{M+1} \\ p_{M+2} \\ \vdots \\ p_N \end{pmatrix}, \quad M \geq 0, \quad M < N.$$

Then, Eq. (1.3) may be divided by the regular operator U from the left, giving $Lz = 0$, i.e.,

$$L\hat{p}z = -L\hat{q}z. \quad (2.7)$$

Next, we use the fact that $\hat{p}L\hat{q} = \hat{p}L\hat{p}\hat{q} = 0$ (identically) and, applying $\hat{p}U\hat{p}$ on both sides of Eq. (2.7), we get

$$(Q^{\text{eff}} - E)\hat{p}z = 0, \quad (2.8)$$

$$Q^{\text{eff}} - E = \hat{p}U\hat{p} \cdot \hat{p}L\hat{p}.$$

This may be interpreted as a Feshbach or Löwdin model-space equation which defines the first $M + 1$ components of the exact wave function coefficients z .

In the $N \rightarrow \infty$ limit, Eq. (2.8) remains valid in the same form, although its derivation must be performed more carefully since the division by U is not permitted anymore.^{6,10} Fortunately, we may understand simply its first M rows as equivalent to the explicit determinantal definitions of p_n 's since $Q_{mn}^{\text{eff}} = Q_{mn}$, for all $n < M$,

$$p_{n+1} = (p_0/B_0 B_1 \cdots B_n) \det(Q^{(n)} - E),$$

$$n = 0, 1, \dots, M-1. \quad (2.9)$$

Then, only the remaining (last) row of Eq. (2.8) [cf. also Eq. (2.3)],

$$Z_M p_M + \sum_{l=1}^q Y_M^{(l)} p_{M-l} + \sum_{l=1}^q X_M^{(l)} p_{M-q-l} = 0, \quad (2.10)$$

has the ECF coefficients that contain all the relevant information about the missing part of the full matrix Q . Their recurrent evaluation is therefore equivalent to a return to the original Eq. (1.3) with $N = \infty$.¹⁴

In essence, our present main purpose lies just in a non-ECF specification of these coefficients. After achieving this purpose [cf. Eq. (3.15) below], we shall be able to treat Eq. (2.10) as a transcendental equation that specifies the binding energies numerically.

III. THE DIFFERENCE-EQUATION CONSTRUCTION OF Q^{eff}

A. The boundary conditions

As shown in Ref. 6, the potentials (1.2) with the "superconfining property"

$$\lim_{r \rightarrow \infty} [V(r) - a_{2q+1} r^{4q+2}] / r^{2q} = +\infty \quad (3.1)$$

enable us to choose the parameters β_k in (1.6) in such a way that all the coefficients $D^{(j)}$ in Q vanish. In the general case, let us now introduce an integer $t \in [0, q]$ such that $D^{(t)} \neq 0$ while $D^{(t+1)} = D^{(t+2)} = \cdots = D^{(q)} = 0$. Then, we may write each row of our infinite-dimensional matrix equation (1.3) as a difference equation for some function h_n ,

$$B_n h_{n+1} - \sum_{j=0}^q C_n^{(j)} h_{n-j} - \sum_{k=1}^t D^{(k)} h_{n-q-k} = 0$$

$$(C_n^{(0)} \equiv A_n - E), \quad (3.2)$$

and put $h_n = p_n$ only after an incorporation of the corresponding boundary conditions.

In general, Eq. (3.2) has a complete set of $q + t + 1$ independent solutions. For example, we may specify $h_n = h_n^{(k)}$, $k = 1, 2, \dots, q + t + 1$, by the set of independent initial values

$$h_{M_k-j}^{(k)} = 0, \quad j = 1, 2, \dots, q + t, \quad (3.3)$$

with some positive integers $M_k \neq M_n$ for $k \neq n$, and write the general solution of (3.2) as a $(q + t)$ -parametric linear superposition

$$h_n = \sum_{k=1}^{q+t+1} \alpha_k h_n^{(k)} \quad (3.4)$$

with an arbitrary normalization. An explicit determinantal representation of the separate components $h_n^{(k)}$ may be derived in full analogy with Eq. (2.9).

According to I, the natural $q + t$ boundary conditions at the origin,

$$h_{-1} = 0, \quad h_{-2} = 0, \dots, h_{-q-t} = 0, \quad (3.5)$$

simply reflect the regularity of $\psi(r)$ at $r \rightarrow 0$. They must be complemented by the physical normalization requirement or boundary condition of the asymptotic HD form¹

$$h_{N+1} = 0, \quad N \rightarrow \infty. \quad (3.6)$$

B. The reverse Hill-determinant method

There is a symmetry between (3.5) and (3.6)—we may restrict (3.4) by $q + t$ conditions (3.5) or by the single requirement (3.6), and obtain a unique "regular" solution or $q + t$ "Jost" solutions, respectively. In the former case, an addition of the asymptotical requirement (3.6) and an availability of the determinantal definition (2.9) lead immediately to the Hill-determinant algorithm. Also, let us study the latter combination of "Jost" solutions with the boundary conditions (3.5).

In the first step, we recall I and replace the complete initializations (3.3) and set of solutions $h_n^{(k)}$ by the other complete set of solutions $h_n^{[s,k]}$ specified by the $q + t + 1$ independent $n \gg 1$ asymptotics

$$h_n^{[d,k]} \approx |(a/n)^{n/(q+1)}| \exp[2\pi k n i / (q+1)],$$

$$n \gg 1, \quad k = 1, 2, \dots, q + 1, \quad (3.7a)$$

and

$$h_n^{[s,k]} \approx |(b/n)^{n/t}| \exp(2\pi k n i / t),$$

$$b = D^{(t)} / 4a, \quad n \gg 1, \quad k = 1, 2, \dots, t. \quad (3.7b)$$

The superscripts d and s mean "dominant" and "subdominant" since

$$(a/n)^{n/(q+1)} / (b/n)^{n/t} = \exp n(\gamma \ln n + \delta),$$

$$\gamma = (q + 1 - t) / t(q + 1) > 0,$$

$$\delta = (\ln a) / (q + 1) - (\ln b) / t, \quad (3.8)$$

increases very quickly with the increasing n . An explicit derivation of the asymptotics (3.7) is easy: we insert the leading-order form of coefficients

$$B_n = 4n^2 + O(n), \quad C_n^{(j)} = 4\beta_{j+1} n + O(1),$$

$$D^{(k)} = O(1) \quad (3.9)$$

into (3.2) and arrive at the two possible leading-order forms of this equation,

$$4n^2 h_{n+1} - 4a n h_{n-q} + \text{corrections} = 0 \quad (3.10a)$$

and

$$4nah_{n-q} - D^{(t)}h_{n-q-t} + \text{corrections} = 0, \quad (3.10b)$$

with the simple solutions (3.7a) and (3.7b), respectively (I).

In the second step, we may incorporate the second-order corrections in (3.10) and find out that the asymptotics of h_n [Eq. (3.4)] are absolutely dominated by a single component $h_n^{[d,q+1]}$. In accordance with the physical boundary condition (3.6), these asymptotics have to change sign at the physical energy E and some large $n = N \gg 1$. Vice versa, the general “Jost” solution $h_N^{(\text{Jost})}$ of (3.2) [compatible with (3.6)] may be characterized simply by an absence of the dominant component $h_N^{[d,q+1]}$,

$$h_N^{(\text{Jost})} = \sum_{k=1}^q \rho_k h_N^{[d,k]} + \sum_{k=1}^t \rho_{k+q} h_N^{[s,k]}. \quad (3.11)$$

In a simplified notation, let us write

$$h_N^{(\text{Jost})} = \sum_{k=1}^{q+t} J_{Nk} \rho_k$$

and define J_{Nk} by Eq. (3.7) in the limit $N \rightarrow \infty$. After an incorporation of a few corrections in (3.10), we also may obtain an asymptotic series and define J_{Nk} for $N \gg N_0 \gg 1$ with a sufficient precision (see below). Then, we recall Eq. (3.2) with indices $n = N_0 + q + t - 1$, $n = N_0 + q + t - 2, \dots, n = q + t$, and use it as a recurrent definition of J_{mk} beyond the asymptotic region, at $m = N_0 - 1, N_0 - 2, \dots, 0$, respectively.

Of course, the remaining $q + t$ rows of Eq. (3.2) [together with the boundary conditions (3.5)] represent simply a $(q + t)$ -dimensional matrix equation

$$\sum_{m=0}^{q+t-1} \left[\sum_{l=0}^{q+t} Q_{kl} J_{lm} - E J_{km} \right] \rho_m = 0, \quad k = 0, 1, \dots, q + t - 1. \quad (3.12)$$

Up to an arbitrary normalization, it fixes the physical energies and coefficients ρ in (3.11)—we may put $p_n = h_n^{(\text{Jost})}$ in (1.5).

C. The general matching conditions

An optimal approximate representation of the wave functions p_n or $\psi(r)$ (1.5) is combined from the $n \gg 1$ asymptotics $p_n = h_n^{(\text{Jost})}$ (3.11) and $n = O(1)$ determinants ($p_n = h_n^{(\text{regular})}$ [Eq. (2.9)], i.e.,

$$h_n^{(\text{regular})} = J_{n0} \rho_0$$

with the single normalization parameter ρ_0). In the numerical practice, the particular boundary-condition matchings (3.11) + (3.5) [i.e., (3.12)] or (2.9) + (3.6) [i.e., (1.8)] may be then replaced also by a more flexible condition $h_{n_i}^{(\text{regular})} = h_{n_i}^{(\text{Jost})}$,

$$J_{M+i,0} \rho_0 = \sum_{k=1}^{q+t} J_{M+i,k} \rho_k, \quad i = 1, 2, \dots, q + t + 1, \quad (3.13)$$

of the “determinant = expansion” type with $M > q + t$.

The necessary dimension of determinants or precision of the expansions may be lowered by a transition to the alternative form of (3.13),

$$\left(\sum_{n=m-q-t}^M Q_{mn} J_{n0} - E J_{M0} \delta_{mM} \right) \rho_0 + \sum_{k=1}^{q+t} \left[\sum_{n=M+1}^{m+1} Q_{mn} J_{nk} - E J_{mk} (1 - \delta_{mM}) \right] \rho_k = 0, \quad m = M, M + 1, \dots, M + q + t. \quad (3.14)$$

This may be understood as a generalization of (3.12). Now, an analogous generalization has to be found for the Hill-determinant prescription (1.8). We shall interpret the “effective-matrix” Schrödinger equation (2.8) also as a matching condition.

In the first step, let us recall the ECF construction of the corresponding “effective” relation (2.10), and notice that its ECF matrix elements Q_{Mm}^{eff} must be independent of the parameters $A_0, A_1, \dots, A_{q+t-1}$. In the present recurrent scheme, we have $J_{n0} = J_{n0}(A_0, A_1, \dots, A_{n-1})$ and $J_{nk} = J_{nk}(A_{n+q+t}, A_{n+q+t+1}, \dots)$ so that $\rho_k = \rho_k(A_0, A_1, \dots, A_{q+t-1})$ after a complete matching (3.14), with any $k \geq 1$. As a result, the matrix elements of Q_{Mm}^{eff} with $M > q + t$ must be also independent of our particular choice of the coefficients ρ_k in the “Jost” solution (3.11) and therefore, its insertion in (2.10) should be valid for all the asymptotic components $h_n^{[x,k]}$ separately,

$$Z_M J_{Mk} + \sum_{m=1}^q Y_M^{(m)} J_{M-mk} + \sum_{m=1}^t X_M^{(m)} J_{M-q-m,k} = 0, \quad k = 1, 2, \dots, q + t. \quad (3.15)$$

This is our main conclusion—we may reinterpret (3.15) as a set of $q + t$ linear homogeneous algebraic equations that define the matrix elements of Q_{Mm}^{eff} as functions of the known (asymptotic-series) matrix elements J_{nk} , $k \geq 1$.

IV. A SIMPLE EXAMPLE

The change of variables in (1.1),

$$\psi \rightarrow r^{\text{const}} \psi, \quad r \rightarrow r^{1/p}, \quad l + \frac{1}{2} \rightarrow (l + \frac{1}{2})/p, \quad p = 1, 2, 3, 4, \quad (4.1)$$

enables us to consider the class of potentials $V = V_p(r)$,

$$\begin{aligned} V_1(r) &= a_1 r^2 + a_2 r^4 + a_3 r^6, \\ V_2(r) &= \frac{1}{2} a_0 / r + 2a_2 r + 4a_3 r^2, \\ V_3(r) &= a_0 / (3r)^{4/3} + a_1 / (3r)^{2/3} + a_3 (3r)^{2/3}, \\ V_4(r) &= \frac{1}{8} a_0 / r^{3/2} + \frac{1}{4} a_1 / r + \frac{1}{2} a_2 / r^{1/2}, \end{aligned} \quad (4.2)$$

as the same eigenvalue problem (1.1) with the energies $E = E_p$,

$$E_1 = -a_0, \quad E_2 = -a_1, \quad E_3 = -a_2, \quad E_4 = -a_3, \quad (4.3)$$

respectively. Thus we may restrict our attention to $p = 1$, $q = 1$ and, inserting the ansatz (1.5) in (1.1) ($\beta_1 = \beta$, $\gamma_1 = \gamma$), obtain the matrix equation (1.3) or difference equation (3.2) with the explicit coefficients

$$\begin{aligned} B_n &= (2n + 2)(2n + 2l + 3), \\ A_n &= C_n^{(0)} - a_0 = (4n + 2l + 3)\beta, \\ C_n &= C_n^{(1)} = (4n + 2l + 1)a + a_1 - \beta^2, \quad n = 0, 1, \dots, \\ D^{(1)} &= D = 2(\gamma - \beta)a, \quad \gamma = a_2/2a > -\beta. \end{aligned} \quad (4.4)$$

From this difference Schrödinger equation, we may return to the various physical situations (4.2) via the transformation (1.5) and (4.1).

A. The asymptotically subdominant solution of the difference Schrödinger equation

Let us complement the explicit coefficients (4.4) by the change of variables

$$h_n = ((\beta - \gamma)/2)^n \varphi_n / n!, \quad n \geq 0, \quad (4.5)$$

and use the notation

$$\dot{\varphi}_n = \varphi_{n+1} - \varphi_n, \quad \ddot{\varphi}_n = \dot{\varphi}_{n+1} - \dot{\varphi}_n. \quad (4.6)$$

We obtain the Schrödinger equation (3.2) in the form

$$\begin{aligned} \dot{\varphi}_{n-1} = (1/n) [\alpha_1 \varphi_n + \alpha_2 \dot{\varphi}_n + \alpha_3 \ddot{\varphi}_n \\ + [1/(n+1)] (\alpha_4 \varphi_{n+1} + \alpha_5 \dot{\varphi}_{n+1})], \end{aligned} \quad (4.7)$$

where

$$\begin{aligned} 4\alpha_1 a &= \gamma^2 - a_1 - (2l+5)a, \\ 4\alpha_2 a &= 2\gamma(\gamma - \beta), \quad 4\alpha_3 a = (\gamma - \beta)^2, \\ 4\alpha_4 a &= [(l + \frac{3}{2})\gamma + a_0/2](\gamma - \beta), \\ 4\alpha_5 a &= (l + \frac{3}{2})(\gamma - \beta)^2. \end{aligned} \quad (4.8)$$

Then, another ansatz

$$\varphi_n = \sum_{k=0}^{\infty} \frac{\Gamma(n+1)}{\Gamma(n+z+k)} c_k \quad (4.9)$$

simplifies the separate differences (4.6)—we may put

$$z = 1 - \alpha_1$$

and get (4.7) in the final form

$$\begin{aligned} (k+1)c_{k+1} - \mu_k c_k - \nu_k c_{k-1} &= 0, \\ \mu_k &= (k - \alpha_1)\alpha_2 - \alpha_4, \\ \nu_k &= (k - \alpha_1 - 1)(\alpha_1\alpha_3 - k\alpha_3 + \alpha_5), \\ c_{-1} &= 0, \quad k = 0, 1, \dots \end{aligned} \quad (4.10)$$

Its algebraic determinantal solution is straightforward,

$$c_{k+1} = \frac{c_0}{(k+1)!} \det \begin{pmatrix} \mu_0 & -1 & 0 & \cdots & 0 \\ \nu_1 & \mu_1 & -1 & 0 & \cdots \\ 0 & 2\nu_2 & \mu_2 & -1 & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & \cdots & 0 & k\nu_k & \mu_k \end{pmatrix}. \quad (4.11)$$

For large k , we have an estimate

$$c_k \approx (\pm (\gamma - \beta)/2 \cdot \sqrt{-k/a})^k \approx k^{k/2}, \quad k \gg 1,$$

which does not contradict the convergence of the ansatz (4.9). Hence we may construct the subdominant solution $h_n^{[s,1]} = h_n^{[s]}$ with an arbitrary precision in principle.

B. The pair of the asymptotically dominant solutions

Let us put $a = \alpha^2$ and not specify the sign of the new parameter α yet. Then, according to I, the alternative ansatz

$$h_n = \left(\frac{\alpha}{\sqrt{2}}\right)^n \frac{\Gamma(C_{n+1}/8a)}{\Gamma(1+n/2)\Gamma[(n+l+\frac{3}{2})/2]} \varphi_n \quad (4.12)$$

converts the Schrödinger equation (3.2) with $q = p = 1$ into the relation (5.8) of I,

$$\begin{aligned} [2 \sinh \frac{1}{2} \nabla + S_n + R_n (\exp \nabla - 1)] \varphi_n &= 0, \\ \nabla &= -2 \frac{d}{dn}, \quad S_n = T_n A_n C_{n-1} / B_{n-1}, \\ R_n &= D T_n, \end{aligned} \quad (4.13)$$

$$T_n = \frac{\sqrt{2} \Gamma[(n+1)/2] \Gamma[(n+l+\frac{3}{2})/2] \Gamma(C_{n-1}/8a)}{(2\alpha)^3 \Gamma(n/2) \Gamma[(n+l+\frac{1}{2})/2] \Gamma(C_{n+2}/8a)},$$

$$n = 0, 1, \dots,$$

where the exponentials $\exp(m/2)\nabla$ are infinite Taylor series representing the shift of indices,

$$\varphi_{n+m} = \sum_{k=0}^{\infty} \frac{m^k}{k!} \frac{d^k}{dn^k} \varphi_n = \exp\left(-\frac{m\nabla}{2}\right) \varphi_n \quad (4.14)$$

and an ansatz has to be used again,

$$\begin{aligned} \varphi_n &= \sum_{k=0}^M b_k \xi_k + O(\xi_{M+1}), \\ \xi_k &= \exp(G\varepsilon^{-1}) \varepsilon^{F+k}, \quad G = (\beta + \gamma)/2\alpha, \\ F &= (\gamma^2 - \beta^2)/4a, \\ \varepsilon &= 1/\sqrt{n}, \quad n \gg 1. \end{aligned} \quad (4.15)$$

Its leading-order coefficients are taken from I, and its general form is dictated by the simple rule for differentiation,

$$\begin{aligned} \nabla^m \xi_k &= \left(\varepsilon^3 \frac{d}{d\varepsilon}\right)^m \xi_k \\ &= \sum_{j=0}^m (-G)^{m-j} W_{j,k}^{(m)} \xi_{k+m+j}, \\ W_{0,k}^{(1)} &= 1, \quad W_{1,k}^{(1)} = k + F, \\ W_{j,k}^{(m+1)} &= W_{j,k}^{(m)} + (F + k + m + j - 1) W_{j-1,k}^{(m)}, \\ j &= 0, 1, \dots, m+1. \end{aligned} \quad (4.16)$$

Obviously, the structure of Eq. (4.13) is much more complicated than the structure of its “subdominantly smooth” counterpart (4.7). Fortunately, the coefficients in (4.13) may be represented by the Barnes formula¹³

$$\begin{aligned} \frac{\Gamma(z+a)}{\Gamma(z+b)} &= z^{a-b} \exp \left[\sum_{m=1}^M z^{-m} \frac{(-1)^{m+1}}{m(m+1)} \{B_{m+1}(a) \right. \\ &\quad \left. - B_{m+1}(b)\} + O(z^{-M-1}) \right], \end{aligned} \quad (4.17)$$

where

$$\begin{aligned} B_2(x) &= x^2 - x + \frac{1}{6}, \\ B_3(x) &= x^3 - 3x^2/2 + x/2, \\ &\vdots \end{aligned}$$

are the so-called Bernoulli polynomials. As a consequence, an $n \gg 1$ asymptotic form of our Eq. (4.13) may again be interpreted as an equation with the power-series coefficients

$$\begin{aligned} T_n &= t_0 \varepsilon + t_1 \varepsilon^3 + \cdots, \quad S_n = s_0 \varepsilon + \cdots, \\ t_0 &= 1/4\alpha^3, \quad s_0 = G_1 \cdots \end{aligned}$$

and schematical structure (n or ε dependence, $\varepsilon \ll 1$)

$$\Theta_1 \varepsilon + \Theta_2 \varepsilon^2 + \Theta_3 \varepsilon^3 + \cdots = 0, \quad (4.18)$$

where, obviously, each coefficient Θ_{M+1} must be equal to zero separately.

In general, the algebraic manipulations represented by Eq. (4.18) are combined from the evaluation of the power series (4.14), the action (4.16), and insertions of (4.17). They are complicated but straightforward; at $M=0$, we obtain an identity due to our choice of $G=s_0$, at $M=1$, we obtain an identity as well, due to our specification of the second free parameter $F=GD_{t_0}$ in (4.15). Finally, it is easy to show that the $M \geq 2$ items of (4.18) do not contain b_M and may be written in the form

$$(M-1)b_{M-1} = \sum_{l=0}^{M-2} b_l \sum_{k_1, k_2, k_3} (-s_{k_1} \delta_{k_2,0} \delta_{k_3,0} + \cdots), \quad (4.19)$$

$$k_1, k_2, k_3 \geq 0, \quad 2k_1 + k_2 + k_3 + l = M.$$

This defines the sequence of the unknown coefficients in our

$$\begin{pmatrix} (A_0 - E) & -B_0 & 0 & \cdots & \cdots \\ C_1 & (A_1 - E) & -B_1 & 0 & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & \cdots & D & C_{N-1} & (A_{N-1} - E) \\ 0 & \cdots & 0 & D & Y_N \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \cdots \\ -B_{N-1} \\ Z_N \end{pmatrix} \begin{pmatrix} p_0 \\ p_1 \\ \cdots \\ p_{N-1} \\ p_N \end{pmatrix} = 0. \quad (4.20)$$

The pair of the unknown matrix elements (normalized by the choice of $X_N^{(1)} = D^{(1)} = D$) $Y_N^{(1)} (= Y_N)$ and Z_N is defined by Eq. (3.15),

$$\begin{aligned} Dh_{N-2}^{[d]} + Y_N h_{N-1}^{[d]} + Z_N h_N^{[d]} &= 0, \\ Dh_{N-2}^{[s]} + Y_N h_{N-1}^{[s]} + Z_N h_N^{[s]} &= 0, \end{aligned} \quad (4.21)$$

with the asymptotic series representations (4.9) and (4.15) of the Jost components $h_n^{[s]}$ and $h_n^{[d]}$, respectively.

In the $N \gg 1$ asymptotic region, we may interpret (4.21) as an estimate

$$\begin{pmatrix} Y_N \\ Z_N \end{pmatrix} = \begin{pmatrix} O(1/\sqrt{N}) & O(1/N) \\ O(1/N) & O(1/N^2) \end{pmatrix}^{-1} \begin{pmatrix} O(1) \\ O(1) \end{pmatrix} = \begin{pmatrix} O(N) \\ O(N^{3/2}) \end{pmatrix}, \quad (4.22)$$

compatible with the ECF results.¹⁶ This estimate enables us to arrange the explicit solution of (4.21) in the form

$$\begin{aligned} Y_N &= \frac{R(s,2) - R(d,2)}{R(s,1) - R(d,1)}, \\ R(x,i) &= \frac{h_{N-i}^{[x]}}{h_N^{[x]}}, \quad x=s,d, \quad i=1,2, \\ Z_N &= \frac{R(s,2)R(d,1) - R(d,2)R(s,1)}{R(s,1) - R(d,1)}. \end{aligned} \quad (4.23)$$

This is our final formula and has the following features.

(a) After an insertion of the expansions of $h_n^{[s]}$ and $h_n^{[d]}$ as derived in the preceding text, this formula is valid for all $N \geq 2$ in principle.

(b) In practice, we shall use the truncated expansions of $h_n^{[x]}$, assuming that the value of the subscript n is sufficiently

ansatz (4.15) in a recurrent way similar to the definition (4.10) of the subdominant solutions.

In the present case, our choice of the sign of α in (4.12) is ambiguous. In accord with I, we have to choose both signs when constructing the general solution. For the elimination of the unphysical asymptotic component we used not changing signs, i.e., with $\alpha > 0$ in (4.12) (see I for more details) and must guarantee its asymptotic dominance [represented by the exponential $\exp G/\varepsilon$ in (4.15), i.e., obviously, just by the condition (1.7)]. Vice versa, the expansions $h_n^{[d]}$ needed in the Jost solution (3.11) will be given by the present formulas with the negative $\alpha < 0$.

C. The matrix elements of the effective Hamiltonian $\mathcal{Q}^{\text{eff}}$

Our basic finite-dimensional and exact form of the Schrödinger equation (2.8) has to be solved numerically, of course. In the present example, it takes the form

large, $n \gg 1$. Then, all the second terms in the numerators and denominators in Eq. (4.23) become mere corrections of $O(1/N^{1/2})$. In the numerator of Y_N , this correction is even smaller [$O(1/N)$].

(c) The Padé-type structure of Eq. (4.23) is suitable for computations even for the intermediate (not too large) values of the subscript N . Vice versa, a power-series rearrangement of Eq. (4.23) could make the N dependence of Y_N and Z_N less complicated in the non-numerical considerations. For this purpose, we also recommend formula (2.5), which defines both Y_N and Z_N as functions of a quantity F_n , $n > N$. In this context, we also recall the results of Ref. 16 where an expansion of the asymptotic series type

$$B_{N-1} F_N = \sqrt{N} \sum_{m=0}^K \delta_m N^{-m/2} \quad (4.24)$$

has been obtained, with the coefficients

$$\begin{aligned} \delta_0 &= 1/\sqrt{a}, \quad \delta_1 = -(\beta + \gamma)/4a, \\ \delta_2 &= (1/16a\sqrt{a})[(4l-2)a + (\beta + \gamma)^2 + 4\gamma^2 - 4a_1], \end{aligned} \quad (4.25)$$

etc. (generated on the computer in REDUCE).

For illustrative purposes, we may recall once more the tests of Tater,¹⁰ who employed the simplest nontrivial example of Ref. 4 and demonstrated the efficiency of the ECF-FP concept as well as an extension of validity of the eigenvalue condition (1.9) numerically. His conclusions were based on the expansions of the power-series type (4.24), but only insignificant deviations appear after a transition to the Padé-type "effective" matrix elements (4.23) in the corresponding secular equation (2.8) or (2.10) in the simple $q=1$

example. Thus we may omit the tables of energies here, reemphasizing once more that equations of the type (4.23) [and (3.15) in general] seem to represent the only feasible non-numerical definition of Q^{eff} for the more complicated ($q > 1$) potentials.

V. SUMMARY

In paper I, the Hill-determinant zeros were shown to coincide with energies in the infinite-dimensional limit only. Now, our main result is the identification of these binding energies with zeros of a finite, $[(M+1) \times (M+1)]$ -dimensional effective Hill determinant $\mathcal{G} = \det(Q^{\text{eff}}(E) - E)$.

This paper remains based on the power series ansatz of I, where an algebraic (determinantal) solvability of the resulting difference Schrödinger equation was employed. Here, we solve it iteratively in the $n \gg 1$ asymptotic domain. This has the following advantages.

(i) In the limit $n \rightarrow \infty$, our difference equation degenerates into a two-term recurrence. Hence, for any potential, it becomes exactly solvable in terms of the gamma functions. This contrasts with the $(q+t+2)$ -term recurrent character of the $n = O(1)$ Hill determinants.

(ii) A transition to large but finite indices $n < \infty$ does not violate the leading-order two-term structure of the equation. A systematic and explicit algebraic evaluation of corrections remains a linear problem to all orders.

(iii) Even in the zeroth-order approximation, we obtain a significant improvement of the Hill-determinant algorithm (cf., e.g., the numerical tests of the particular examples in Refs. 6 and 10). In the present formalism, the reasons are

obvious—we have chosen a better matching point for the “regular” and “Jost” solutions.

(iv) The implicit Padé-approximation-like structure of our basic formulas [(3.15) or, in an explicit example, (4.23)] seems promising for an *a priori* control of precision at large $M \ll \infty$.

(v) In contrast to the geometric (fixed-point) analysis of convergence (related to the ECF definition of Q^{eff} and given in detail in Refs. 6 and 16), our present definitions [containing complex roots (3.7), etc.] seem more likely to clarify the origin and elimination of oscillations as observed in Ref. 14.

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